Bridging Life Cycle Assessment and orbital sustainability science: a multi-domain composite indicator for the environmental assessment of space missions

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Abstract

The environmental footprint of a space mission spans three domains presently assessed in isolation: the terrestrial life cycle of manufacturing and ground operations, the atmospheric perturbation of launch and re-entry, and the orbital impact of debris and congestion. This paper introduces the Space Sustainability Composite Indicator (SSCI), a composite sustainability indicator coupling terrestrial Life Cycle Assessment (EF 3.1) with orbital and atmospheric metrics. For the orbital domain it adopts an ECOB-aligned debris proxy and adds two complementary categories, operational congestion and in-orbit material criticality, each normalised to a reference mission before aggregation, plus stratospheric factors for launch and re-entry. Linear and geometric-mean aggregations are compared, with robustness to the weighting, stratospheric factors and normalisation reference quantified. The framework is released as an open toolchain coupling OpenLCA, Python and a MATLAB orbital module, with a 54-process inventory library. Three case studies (a PocketCube, Sentinel-6 and the heritage ENVISAT platform) anchor three regimes in which congestion, material criticality and debris generation respectively dominate the orbital score, a ranking invariant to the normalisation. Scaling to 300 missions drawn from the real active-LEO altitude distribution shows that most occupy shells far denser than a representative reference orbit, so their absolute congestion burden, invisible to terrestrial LCA, is first-order; a terrestrial-only PEFCR4Space-style indicator therefore underestimates the multi-domain footprint for the large majority of missions. Congestion depends on orbital presence, not spacecraft mass, so small satellites in contested shells are not environmentally negligible.

Keywords: space sustainability, Life Cycle Assessment, composite indicator, orbital debris, launch emissions, PEFCR

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Nomenclature

$\tilde{I}_T, \tilde{I}_A, \tilde{I}_O$	normalised terrestrial, atmospheric, orbital impact scores
$SSCI_m$	composite indicator of mission m
w_T, w_A, w_O	domain weights ($w_T + w_A + w_O = 1$)
DGP	debris-generation potential (ECOB, proxy in pilot)
CC	congestion contribution
MC	in-orbit material criticality
V_m^{occ}	operational exclusion volume
ρ_{op}	operational population density
T_{op}	operational lifetime
q_i, χ_i	mass and terrestrial criticality of material i
η_i^{orbit}	orbit-loss multiplier
κ_e, τ_e	effectiveness coefficient and stratospheric residence time of species e
$m_{e,m}$	mass of species e emitted by mission m

1. Introduction

The past five years have seen the number of operational satellites in Low Earth Orbit (LEO) more than triple, with current trajectories pointing toward over 100,000 active spacecraft by 2030 (European Space Agency, 2024). Commercial mega-constellations, lower launch costs, miniaturisation and the expansion of state and private actors have reshaped the orbital environment from a lightly used scientific domain into a crowded commercial one.

This expansion is generating environmental pressures across three distinct domains. On the surface and within the atmosphere, the manufacturing, propellant production, ground operations and disposal of spacecraft contribute to the impact categories tracked by conventional industrial Life Cycle Assessment (LCA): greenhouse gas emissions, resource depletion, toxicity, eutrophication. In the upper atmosphere, the increasing frequency of launches and re-entries introduces emissions whose stratospheric residence times and radiative behaviour differ substantially from surface-level analogues, with documented potential to alter ozone chemistry and aerosol loading (Ross and Toohey, 2019; Maloney et al., 2022). In the orbital domain, the accumulation of debris, the propagation of collision risk and the growing congestion of preferred regions represent an unprecedented form of degradation that imposes long-term constraints on access to the resource itself.

The response to this crisis has so far been fragmented across disciplines. Within the Life Cycle Assessment community, a body of work has emerged over the past decade to adapt the ISO 14040/14044 methodological framework to the specific characteristics of space systems. Pioneering contributions (Maury et al., 2017; Ross and Toohey, 2019; Maury et al., 2020; Dallas et al., 2020; Wilson and Vasile, 2023) established the feasibility of applying LCA to spacecraft manufacturing, launch operations and end-of-life disposal, and quantified the relative contributions of different mission phases to terrestrial impact categories. This trajectory has culminated in the publication, in 2025, of the first draft of the Product Environmental Footprint Category Rules for Space (PEFCR4Space), developed under the auspices of the European Space Agency and the European Commission Joint Research Centre, and currently under open consultation (European Commission and European Space Agency, 2025). PEFCR4Space represents a substantial methodological consolidation: it standardises product categories, functional units, system boundaries and characterisation factors for the manufacturing and ground phases of a space mission, and provides the first regulatory-grade reference for comparative environmental claims in the space sector.

In parallel, and largely independently, the orbital sustainability research community has developed a separate family of metrics aimed at quantifying the environmental impact of spacecraft on the orbital environment itself. The Environmental Consequences of Orbital Breakups (ECOB) index, introduced by Letizia, Colombo, Lewis and Krag (Letizia et al., 2016, 2017), remains the most consolidated framework in this area, capturing the contribution of an individual mission to long-term debris flux and to the integrated population risk for other operational spacecraft. ECOB-derived metrics have been progressively extended to account for fragmentation severity, breakup geometry, and altitude-dependent residence times, and are increasingly cited in policy documents on debris mitigation and on the implementation of Zero Debris frameworks (European Space Agency, 2023; United Nations Committee on the Peaceful Uses of Outer Space, 2019).

A third family of methods, rooted in classical engineering risk assessment (European Cooperation for Space Standardization, 2008; National Aeronautics and Space Administration, 2019; Toson, 2026), formalises the probabilistic treatment of mission-level hazards through risk indices that combine probability of occurrence with severity of consequence. These methods have been refined for collision risk classification at both individual and collective levels, but they have not, until now, been integrated into a sustainability assessment framework.

The result is a patchwork: LCA captures terrestrial and atmospheric impacts but stops at the launch event; orbital sustainability indices capture in-orbit behaviour but presuppose the lifecycle accounting they do not perform; and risk-based engineering methods quantify mission-specific consequences but do not aggregate across missions or domains. These three perspectives have not yet been integrated within a single quantitative framework spanning the entire mission lifecycle.

This work addresses a question that, despite the rapid maturation of each of the three contributing fields, currently has no consensus answer in the literature: *how can the environmental impact of a space mission be quantified, compared and predicted across all relevant domains and lifecycle phases, within a single composite metric?* This paper proposes the **Space Sustainability Composite Indicator (SSCI)**, a multi-domain Life Cycle Assessment framework that integrates terrestrial, atmospheric and orbital impacts of space missions into a single composite quantification. The SSCI is built upon, and extends, the EF 3.1 characterisation method. For the orbital domain it adopts the established ECOB index (Letizia et al., 2016, 2017) as its debris-impact component instead of re-deriving a debris metric, and adds two complementary categories that ECOB does not address (orbital congestion during the operational phase and in-orbit material criticality). It further incorporates atmospheric perturbation models calibrated for launch and re-entry phases, and adopts a risk-based aggregation rationale derived from established engineering practice (Toson, 2026).

This work makes five contributions. It formalises a multi-domain composite indicator for the sustainability of space activities, coupling terrestrial Life Cycle Assessment, atmospheric science and orbital sustainability. An ECOB-aligned debris characterisation is integrated into the framework and complemented by two categories ECOB does not address, operational congestion and in-orbit material criticality. The SSCI is defined as an integrated metric with alternative aggregation and weighting strategies and a robustness analysis. The framework is released as an open toolchain coupling OpenLCA, Python and a MATLAB orbital module, with the SusLifePath library of 54 audited space-sector unit processes. Finally, the regulatory interfaces relevant to European space-sustainability governance are mapped.

Three case studies spanning more than four orders of magnitude in dry mass, a 2P PocketCube, the Sentinel-6 Michael Freilich smallsat and the heritage ENVISAT platform, together with a 300-mission catalogue-scale sample, demonstrate the discriminatory power of the framework and quantify the systematic gap between the composite SSCI and the single-indicator approaches currently in use.

The remainder of the paper reviews the state of the art across the three contributing disciplines (Section 2), develops the conceptual and mathematical framework and its toolchain (Section 3), presents the pilot application with sensitivity and catalogue-scale analyses (Section 4), discusses the methodological, regulatory and policy implications (Section 5) and concludes (Section 6).

2. State of the art

The three traditions this work unifies (Life Cycle Assessment of space systems, orbital sustainability indices and engineering risk assessment) have developed along parallel tracks since the early 2010s. Each is reviewed below for what prevents it from independently supporting a comprehensive sustainability assessment; Section 2.4 articulates the integrated gap.

2.1. Life Cycle Assessment of space systems

The application of LCA to space activities began in the early 2010s, as the rapid growth of the launch market raised the prospect of environmental accountability comparable to terrestrial industrial sectors. Early contributions adapted ISO 14040/14044 principles (International Organization for Standardization, 2006a,b) to spacecraft manufacturing inventories and screened dominant impact categories for representative platforms (Maury et al., 2017; Wilson and Vasile, 2023).

A second wave, from around 2017, broadened the scope to launch operations, ground segments and end-of-life disposal (Maury et al., 2017, 2020), establishing that manufacturing emissions do not always dominate: depending on the impact category, propellant production and the launch event can contribute a comparable or larger share. A parallel line at the JRC and ESA Clean Space Office formalised the choices needed for space-specific characteristics such as critical materials, non-renewable launch fuels and the disposition of upper stages (European Space Agency, 2016).

A third and recent wave of contributions has addressed the atmospheric dimension of space operations. Ross and Toohey (2019) quantified the projected stratospheric emissions of a fully realised mega-constellation scenario, including nitrogen oxides, alumina particulates and black carbon, and flagged concerns for ozone chemistry and stratospheric aerosol loading. Dallas et al. (2020) provided the first systematic review of launch-related emissions within an LCA framework. Subsequent atmospheric modelling (Maloney et al., 2022; Ryan et al., 2022; Ferreira et al., 2024) refined the residence-time estimates of re-entry oxidation products and informed regulatory discussion of high-cadence launch operations.

Standardisation is now in progress. The ESA Space System LCA Guidelines (European Space Agency, 2016) consolidated the choices for ground and launch phases, and the Product Environmental Footprint Category Rules for Space (PEFCR4Space), released as a first draft in 2025 under open consultation (European Commission and European Space Agency, 2025), are the most advanced effort to date toward a regulatory-grade methodology for the environmental certification of space products. The draft defines product categories, boundaries, functional units and characterisation factors compatible with EF 3.1 (Fazio et al., 2018).

Despite this consolidation, gaps remain. PEFCR4Space does not provide formal accounting for orbital debris generation, in-orbit congestion or the criticality of materials used in operational orbits; atmospheric perturbations are recognised as relevant but not yet characterised with the maturity of terrestrial categories; and cross-mission comparability is limited by the strong dependence on the chosen system boundary and functional unit. These gaps, summarised in Table 1, motivate the present work jointly with the limitations of orbital sustainability indices.

2.2. Orbital sustainability indices

The need to quantify the environmental impact of a spacecraft on the orbital environment itself, distinct from its terrestrial and atmospheric impacts, emerged in the late 2000s as a response to the rapid growth of the resident debris population. Among the proposed metrics, the ECOB index (Letizia et al., 2016, 2017) has become the most widely adopted reference. ECOB captures the expected long-term contribution of an individual mission to the integrated fragment population, weighted by the resulting collision risk for other operational spacecraft. Its definition relies on the propagation of representative fragment cloud distributions through long-term debris evolution models and on the integration of the resulting flux against the operational population of target altitudes.

Subsequent contributions have extended ECOB to account for fragmentation severity and breakup geometry (Letizia et al., 2019), for the operationalisation of environment capacity as an early-design driver (Letizia and Lemmens, 2020), and for the explicit treatment of altitude-dependent residence times. Variants and parallel formulations include the Criticality of Spacecraft Index of Rossi et al. (2015) and a family of debris environment ratings used in policy and insurance contexts, including the ranking of the fifty statistically-most-concerning derelict objects in LEO (McKnight et al., 2021). These indices share a common conceptual structure: they quantify the contribution of a single mission to a collective long-term degradation of a shared resource, the orbital environment.

The strength of orbital sustainability indices lies in their physical grounding: they capture debris dynamics, fragmentation geometry and the long-term consequences of operational choices in a way no terrestrial LCA can replicate. As standalone sustainability metrics, however, they have three limitations. They operate independently of any lifecycle accounting, so a spacecraft optimised for low ECOB but built with carbon-intensive processes may still carry a high overall footprint. They typically address a single dimension of orbital impact, debris generation, neglecting operational congestion and the depletion of critical orbital regimes. And they presuppose the inventory data and impact characterisation that LCA produces without performing that accounting themselves. Alone, they cannot support a holistic sustainability assessment.

2.3. Risk-based engineering approaches

A third methodological tradition, rooted in classical engineering risk assessment, treats environmental and operational hazards as quantifiable through the combination of probability and severity. The European Cooperation for Space Standardization framework (European Cooperation for Space Standardization, 2008) codifies this approach for the entire lifecycle of a space project, defining likelihood and severity scoring schemes that are then combined into a risk index. The corresponding NASA framework (National Aeronautics and Space Administration, 2019) operates on equivalent principles. These frameworks are extensively applied to mission design, insurance pricing and project management, but their application to environmental impact assessment has remained limited until recently.

A recent extension of risk-based reasoning to satellite collision classification (Toson, 2026), building on earlier collective and individual risk formulations (Toson, 2022), combines a probabilistic treatment of the orbital event with a severity model based on the projected consequences of fragmentation, producing a continuous risk index for mission classification. The same probability–severity structure transposes to sustainability assessment and provides the scaffold connecting this work to consolidated engineering practice.

The limitation of risk-based approaches for sustainability assessment lies in their scope: designed to quantify hazard for an individual mission and event, they do not natively aggregate across missions, connect to lifecycle inventories or produce composite indicators. Their value here lies in the rigour of their probabilistic foundations, which inform the aggregation rationale of the SSCI.

2.4. Synthesis: the unifying gap

The three traditions reviewed above have each matured in isolation. Table 1 summarises the coverage of each tradition against the eight dimensions that, taken together, constitute the requirements for a comprehensive sustainability assessment of space activities.

The pattern in Table 1 is structural, not incidental: no existing methodology addresses more than half of the required dimensions, and those systematically left uncovered, in particular the integration of orbital impact within a lifecycle framework and the production of a single composite indicator, are the ones that regulatory and design decision-makers increasingly demand.

The integrated gap is not the absence of one methodology but of a framework recognising that these traditions all describe aspects of the same process, the lifecycle of a space mission. Section 3 takes this as its starting point.

Table 1: Coverage of existing methodological traditions against the requirements for a comprehensive multi-domain sustainability assessment of space missions. ● fully addressed, ◐ partially addressed, ○ not addressed.

Dimension	ESA LCA	PEFCR	ECOB	ECSS	**SSCI**
Terrestrial impact (manufacturing, ground)	◐	●	○	○	●
Atmospheric impact (launch, re-entry)	◐	◐	○	○	●
Orbital debris generation	○	○	●	◐	●
Orbital congestion contribution	○	○	◐	○	●
In-orbit material criticality	○	○	○	○	●
Risk-based aggregation	○	○	◐	●	●
Single composite indicator	○	◐	○	◐	●
Open computational toolchain	○	◐	○	○	●

3. Methodological framework

3.1. Conceptual model: the three domains

The environmental footprint of a space mission cannot be reduced to a single physical or temporal scale. Three distinct domains of impact are identified, each with its own physical processes, temporal horizon and relevant impact categories.

The **terrestrial domain** (T) encompasses all environmental exchanges that occur on the surface of the Earth or within the troposphere, from the extraction of raw materials and the manufacturing of the spacecraft to the integration of the platform, ground operations, and the recycling or disposal of recoverable elements at end of life. The relevant impact categories in this domain are those of the EF 3.1 method (Fazio et al., 2018), with inventory data adapted to space-specific processes through the SusLifePath process library described in Section 3.5.

The **atmospheric domain** (A) captures the exchanges during ascent through the upper troposphere and stratosphere and during re-entry of spacecraft and upper stages. Its species (nitrogen oxides, alumina particulates, black carbon, water vapour and chlorine-bearing compounds) have stratospheric residence times of months to years, and their high-altitude behaviour is not adequately described by terrestrial characterisation factors (Ross and Toohey, 2019; Maloney et al., 2022).

The **orbital domain** (O) captures the exchanges during the operational lifetime and post-mission disposal: the contribution to long-term debris accumulation, the congestion of operationally important regions and the depletion of critical orbital regimes. Its categories (Section 3.2) integrate debris environment models with concepts of material criticality from terrestrial industrial ecology.

The three domains are not disjoint: a launch event generates terrestrial emissions through propellant manufacturing, atmospheric emissions through combustion and upper-stage re-entry, and orbital effects through insertion into a contested region. An explicit boundary convention keeps these separate. Combustion and re-entry emissions are accounted in the atmospheric domain, while the terrestrial product systems are built from the spacecraft material inventory and assembly energy and do not yet include propellant manufacturing. There is therefore no double counting between the two domains, and propellant production is a known omission of the terrestrial inventory

(Section 4.1). Figure 1 summarises the structure and the normalised aggregation into the composite.

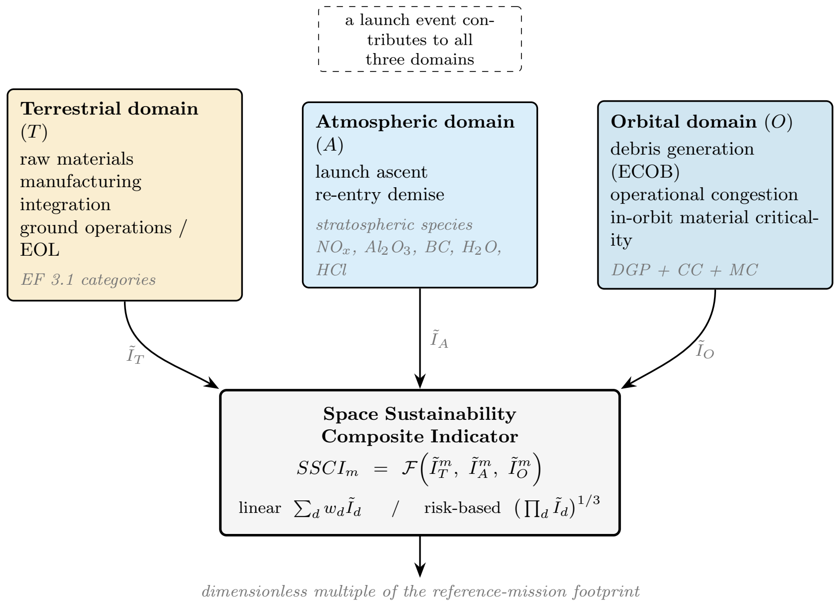

Figure 1: Three-domain conceptual model of the SSCI. Each domain carries its own lifecycle phases, characterisation basis and temporal horizon; the three normalised domain scores $\tilde{I}_T$, $\tilde{I}_A$, $\tilde{I}_O$ are aggregated into the composite indicator under either a linear or a risk-based rule. A single launch event contributes to all three domains.

3.2. The orbital domain: adoption of ECOB and two complementary categories

A central design choice is that the SSCI does **not** reinvent the quantification of orbital debris impact. The ECOB index (Letizia et al., 2016, 2017) is a mature, validated and widely adopted formulation, grounded in long-term debris-evolution modelling, and is the appropriate state-of-the-art measure of a mission's contribution to the resident debris population. The orbital contribution of this work is therefore two-fold: ECOB is adopted as the debris component, integrated for the first time within a lifecycle-assessment framework; and two complementary categories ECOB does not address are introduced—orbital congestion during the operational phase and the criticality of materials lost to orbit. The orbital score $\tilde{I}_O$ is a normalised combination of the three; the three are complementary by regime (each dominates a different mission class, Section 4.2), not by statistical independence.

Debris generation. The debris-generation potential (DGP) is, by definition, the ECOB debris-environment characterisation of a mission's long-term contribution to the resident debris population, weighted by the resulting collision risk for the operational population:

$$DGP_m \equiv \mathrm{ECOB}_m. \tag{1}$$

Conceptually *DGP* is the full ECOB index, computed through the long-term debris-evolution methodology of Letizia et al. (2016, 2017); Letizia and Lemmens (2020) over the 200-year horizon standard in that literature. The results reported here, however, are produced by a transparent ECOB-aligned proxy from the companion engineering paper (Toson, 2026). For a target mission the proxy combines the local object density, the average relative velocity, the mission's cross-sectional area and its residence time into a snapshot collision flux, weighted by a mass-scaled fragmentation-severity term, yielding a per-mission collision-risk number on the ECOB ranking scale. It reproduces the ECOB ranking of the case studies and correlates with an ECOB-aligned formulation at rank correlation 0.91 over a 103-object sample. The proxy is a scope choice for the pilot: it avoids a full debris-evolution (MASTER/DRAMA-class) run while preserving the ranking, and the framework accepts the full ECOB index in its place without any change to the rest of the structure. Adopting ECOB instead of re-deriving a debris metric concentrates the novelty of this work on the multi-domain integration and the two complementary categories.

Congestion contribution. The congestion contribution (*CC*) quantifies a mission's contribution to the operational saturation of its insertion orbit during its active lifetime, distinct from the long-term debris dynamics of *DGP*. Even nominally compliant missions, with adequate post-mission disposal, contribute to the congestion of valuable orbital regions during their service. The *CC* score is defined as

$$CC_m = \int_{t_0}^{t_0+T_{op}} V_m^{occ}(h_m, t)\, \rho_{op}(h_m, t)\, dt, \tag{2}$$

where V_m^{occ} is the operational exclusion volume associated with the mission, T_{op} is the operational lifetime, and the remaining symbols carry the same meaning as in Equation (1). The exclusion volume is taken as the $(50\text{ km})^3$ keep-out box of conjunction (close-approach) analysis (European Cooperation for Space Standardization, 2024), a screening convention set by tracking uncertainty rather than object size.

The distinction between *CC* and the debris component matters, since the two could superficially read as the same density integral. They differ in time window, affected population and nature of the impact. The debris component characterises what a mission may leave behind: it integrates the consequences of a potential fragmentation against the resident population over a multi-decade horizon, and is dominated by the platform's post-operational fate: disposal compliance and residual lifetime. The congestion contribution characterises what the mission certainly occupies while operating: the exclusion volume that conjunction screening must protect against the *active* population, over the operational lifetime only, with no fragmentation needed for the impact to materialise. The two are therefore disjoint in time (intra- versus post-T_{op}) and address different harms (operational traffic burden versus long-term debris hazard).

This makes *CC* an accounting of operational screening burden, not a generic density metric. A fully compliant mission with negligible debris signature still consumes orbital carrying capacity during service: every occupied volume-year in a dense shell translates into conjunction screenings, collision-avoidance manoeuvre budget and coordination overhead for all co-located operators, a load that grows with traffic density and is documented as a first-order trend in current environment reporting (European Space

Agency, 2024). Anchoring V_m^{occ} to the keep-out convention of the ECSS conjunction-analysis standard (European Cooperation for Space Standardization, 2024) ties the category to that screening practice and gives CC the interpretation of satellite-year exposure within protected conjunction volumes.

A consequence is that CC does not depend on spacecraft size. The conjunction keep-out box is set by orbit-determination uncertainty and the screening miss-distance threshold, governed by tracking and ephemeris quality and not by the object's physical dimensions, so a CubeSat and a large platform sharing an altitude impose a comparable screening burden; since orbit-determination uncertainty is typically larger for small, poorly tracked objects, a size-independent keep-out is, if anything, conservative for the small-satellite case. CC therefore measures congestion as a function of orbital *presence*, not mass: a small satellite in a contested shell is not, on this axis, environmentally negligible. This size-independence is a property of the screening convention adopted here, presented as a policy-relevant lens; per-operator thresholds may scale weakly with object class, attenuating but not removing the effect, since orbit-determination uncertainty runs the opposite way for poorly tracked small objects. The active population is concentrated in the most congested shells, which is what makes the catalogue-scale behaviour of Section 4.5 non-trivial.

In-orbit material criticality. The in-orbit material criticality (MC) quantifies a mission's contribution to the depletion of materials whose recoverability or substitutability is degraded by orbital deployment. It integrates material-criticality concepts from terrestrial industrial ecology (Graedel et al., 2015) with the recognition that loss of access to materials in orbit, irrespective of their terrestrial abundance, is a form of environmental degradation specific to the space domain. The MC score is defined as

$$MC_m = \sum_{i \in M_m} q_i \chi_i \eta_i^{orbit}, \tag{3}$$

where M_m is the set of materials present in the spacecraft, q_i is the mass of material i, χ_i is the terrestrial criticality factor of material i (Graedel et al., 2015), and η_i^{orbit} is an orbit-loss multiplier that captures the irreversibility of orbital deployment for the specific material.

The orbit-loss multiplier distinguishes in-orbit material criticality from its terrestrial counterpart and requires justification. It is anchored to two physical regimes. Below approximately 600 km, atmospheric drag returns deployed material within years, consistent with the post-mission disposal rules (International Organization for Standardization, 2019; European Cooperation for Space Standardization, 2024): the material is lost to the terrestrial circular economy for the mission duration and destroyed on demise (burn-up during atmospheric re-entry), but the orbital reservoir clears on an industrial planning horizon, so η^{orbit} takes a floor of 0.3 rather than zero. Above approximately 2000 km, passive decay exceeds centuries and deployment is permanent for any practical purpose, so η^{orbit} saturates at unity. Between the anchors it is interpolated linearly. The linear form is a parsimonious first-order choice: the physical content lies in the two anchors, which reflect the steep growth of residual lifetime with altitude (Toson, 2026), and any monotonic interpolation preserves the ranking. A derivation of η^{orbit} from decay physics and criticality assessment (Graedel et al., 2015) is left to future work.

The mass-weighted formulation has one limitation: because each material enters MC through its mass q_i, the category understates uses in which a highly critical element is present at trace mass. The pilot offers a concrete example: the PocketCube optical payload carries indium-tin-oxide coated glass, in which indium, a critical raw material in the EU assessment, is present at a fraction of a gram. Its mass-weighted score is negligible even at maximum criticality factor, although the function it enables is irreplaceable. Mass weighting is retained because it keeps the category commensurable with the inventory-based structure of the framework, but a complementary per-element scarcity weighting is a natural refinement, and the indium example marks the boundary of validity.

The adopted ECOB index and the two complementary categories are combined into the orbital impact score by normalising *each* sub-category to the reference mission before aggregating,

$$\tilde{I}_O^m = \tfrac{1}{3}\left(\frac{DGP_m}{DGP_{ref}} + \frac{CC_m}{CC_{ref}} + \frac{MC_m}{MC_{ref}}\right), \tag{4}$$

so that the three sub-categories enter on a common dimensionless scale. Per-sub-category normalisation is necessary because DGP, CC and MC carry different units (a normalised collision-risk ratio, a satellite-year exposure and a mass-weighted criticality); summing them in native units would let whichever has the largest numerical magnitude dominate irrespective of physics. The reference is itself a modelling choice: since each sub-category is divided by the reference mission's value of that sub-category, which sub-category *dominates* a mission's orbital score is measured relative to the reference and is therefore reference-dependent, a sensitivity quantified in Section 4.5 (Table 6); the mission ranking and each case study's dominant sub-category are, by contrast, invariant to the reference. The novelty of the orbital domain lies not in the debris metric (the ECOB-aligned characterisation of Section 3.2) but in its integration within a multi-domain framework and in the congestion and material-criticality dimensions ECOB does not capture.

3.3. Atmospheric perturbation modelling

The atmospheric domain requires characterisation factors that are calibrated for stratospheric processes and that cannot be reduced to the troposphere-based factors used in the EF 3.1 method. The framework adopts a three-step approach.

First, the launch and re-entry inventory is extended to species relevant for stratospheric chemistry: nitrogen oxides (NO_x), alumina particulates (Al_2O_3), black carbon (BC), water vapour (H_2O) and chlorine-bearing species (HCl), from launch vehicle propellant chemistry and re-entry oxidation models (Ross and Toohey, 2019; Ryan et al., 2022; Ferreira et al., 2024), with residence times τ_e from the recent ozone assessment literature (World Meteorological Organization, 2022).

Second, the characterisation factors are derived as the product of a radiative or chemical effectiveness coefficient κ_e and the residence time τ_e. The atmospheric impact score for a mission is then

$$\tilde{I}_A = \sum_{e \in E_{atm}} \kappa_e \, \tau_e \, m_{e,m}, \tag{5}$$

where $m_{e,m}$ is the total mass of species e emitted by mission m during ascent and re-entry, and E_{atm} is the set of stratospheric species considered. The ordering of the coefficients κ_e follows the WMO 2022 assessment of spaceflight influences on stratospheric ozone (World Meteorological Organization, 2022, Sec. 7.2.8.1) and the anchors of the recent modelling literature (Maloney et al., 2022; Ryan et al., 2022; Ferreira et al., 2024); the residual uncertainty on the alumina and black-carbon coefficients, which WMO flags as unresolved, is treated in Section 4.3.

Third, the score is normalised against the same reference mission (the 500 kg kerosene-LOX smallsat, Section 3.4). Because the reference is kerosene-fuelled, a solid-rocket-motor launch is scored relative to a clean-launch baseline; the heritage platform's large normalised atmospheric score is therefore partly a property of that reference choice, whereas the ranking of the raw atmospheric burdens is reference-independent.

3.4. The SSCI: mathematical formulation

The Space Sustainability Composite Indicator combines the three domain-specific scores into a single composite quantification. The general formulation is

$$SSCI_m = \mathcal{F}\Big(\tilde{I}_T^m,\ \tilde{I}_A^m,\ \tilde{I}_O^m\Big), \tag{6}$$

where $\tilde{I}_T^m$, $\tilde{I}_A^m$ and $\tilde{I}_O^m$ are the normalised impact scores of mission m in the terrestrial, atmospheric and orbital domains respectively, and $\mathcal{F}$ is the aggregation function. Two aggregation strategies are developed and compared.

The baseline formulation is a weighted linear combination,

$$SSCI_m^{lin} = w_T\,\tilde{I}_T^m + w_A\,\tilde{I}_A^m + w_O\,\tilde{I}_O^m, \quad w_T + w_A + w_O = 1, \tag{7}$$

where the weights w_T, w_A and w_O reflect the relative policy importance assigned to each domain. Following standard composite-indicator practice (OECD and European Commission Joint Research Centre, 2008), three weighting schemes are provided: equal weights ($w_T = w_A = w_O = 1/3$), an illustrative governance-maturity scenario ($w_A = 0.40$, $w_O = 0.35$, $w_T = 0.25$), and an Analytic Hierarchy Process option (Saaty, 1980) that would derive the weights from a pairwise comparison matrix and report its consistency ratio; no AHP elicitation is performed here, the option being provided as the entry point for structured stakeholder elicitation. The governance-maturity ordering reflects how completely each domain is regulated rather than its absolute importance: terrestrial impacts are covered by mature regulation and mitigation supply chains; orbital impacts are partially governed through debris-mitigation standards and emerging traffic management; stratospheric perturbations are subject to no binding instrument and so have the fewest safeguards against accumulation. The scheme is illustrative pending a documented elicitation, and Section 4 quantifies, through Dirichlet sampling of the weight simplex, how much of the composite depends on the weighting.

The linear rule is fully compensatory: a low score in one domain offsets a high score in another. A second formulation aggregates the three domain scores through their geometric mean,

$$SSCI_m^{geo} = \left(\prod_{d\in\{T,A,O\}} \tilde{I}_d^m\right)^{1/3}. \tag{8}$$

The geometric mean has the opposite sensitivity to the arithmetic: bounded above by the linear score, pulled toward the smallest domain score, and collapsing toward zero if any domain score is near zero. The two are therefore reported side by side, not reduced to one normative choice—the arithmetic emphasises the largest domain, the geometric the smallest. Their ratio $SSCI^{lin}/SSCI^{geo}$ grows with the inter-domain spread and serves in Section 4.2 as a diagnostic of how unevenly a mission's impact is distributed.

Normalisation is performed against the reference mission, a notional 500 kg smallsat in a 700 km sun-synchronous orbit (the same reference as the companion engineering paper (Toson, 2026), specified in Section 4.1), so the SSCI is interpretable as a dimensionless multiplier of the reference-mission footprint.

3.5. Toolchain implementation

The framework is implemented as an open toolchain coordinated by a Python orchestrator that reads mission descriptors in a structured YAML schema and produces the domain scores, the composite SSCI and the sensitivity diagnostics. The terrestrial domain is computed in OpenLCA (GreenDelta GmbH, 2024) on a database combining the EF reference data and the official EF 3.1 implementation with the SusLifePath library, 54 audited space-sector unit processes organised in seven lifecycle tiers (raw materials, components, integration, ground operations, launch terrestrial share, operations support, end of life); product systems are built programmatically from each mission's material inventory, with a documented assembly-integration-test electricity overhead, and the EF 3.1 weighted single score is retrieved through the openLCA inter-process interface. The atmospheric domain is a dedicated Python module implementing the characterisation of Section 3.3, with emission indices resolved per propellant chemistry. The orbital domain combines the debris component (the pilot snapshot proxy of Section 3.2) and the congestion module, both executed in MATLAB through a subprocess bridge against the Celestrak snapshot, with the material-criticality module in pure Python. The toolchain is released under MIT licence and deposited on Zenodo. The architecture is shown in Figure 2.

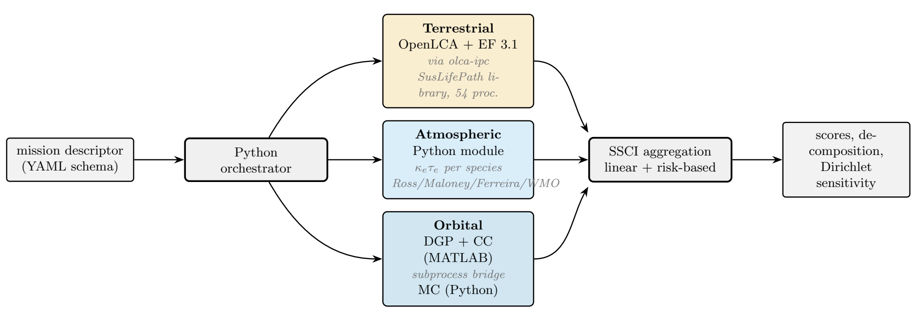

Figure 2: Computational toolchain of the SSCI framework. A Python orchestrator reads the mission descriptor and dispatches the three domain modules: the terrestrial EF 3.1 score is computed in OpenLCA through the olca-ipc bridge on the SusLifePath library, the atmospheric score in a dedicated Python module, and the orbital score through a MATLAB subprocess bridge (debris and congestion) coupled to a Python material-criticality module. The domain scores are aggregated, under both rules, into the composite and its sensitivity diagnostics. The toolchain is released under MIT licence and archived on Zenodo.

Table 2: Three case studies for the SSCI pilot.

Case	Class	Dry mass	Altitude	Launcher	Status
RedPill / J2050	2P PocketCube	0.435 kg	550 km SSO	Falcon 9 rideshare	future
Sentinel-6 M. F.	Medium smallsat	1192 kg	1336 km	Falcon 9 dedicated	operational
ENVISAT	Large heritage	8211 kg	770 km SSO	Ariane 5 G+	defunct, 2012

4. Pilot application

The SSCI is applied to three missions from distinct classes: a 2P PocketCube (a sub-kilogram CubeSat-class platform), the medium-class Sentinel-6 Michael Freilich smallsat and the large legacy ENVISAT platform. They span more than four orders of magnitude in dry mass and use three different launchers, exposing the framework to a diverse set of inventory inputs while keeping the inventory within publicly documented data.

4.1. Case studies and inventory

The three case studies are summarised in Table 2. The selection is driven by three principles: representativeness of the active LEO population, availability of public-domain inventory data, and a span of orbital and propulsion characteristics sufficient to expose the framework's discriminatory power.

The PocketCube class (Porcarelli et al., 2026, 2025a,b; Vignato et al., 2025) has the lowest absolute footprint per mission but the highest growth rate; Sentinel-6 represents the modern scientific smallsat with controlled disposal and clean launcher chemistry; ENVISAT is the upper baseline of the debris-hazard literature (Letizia et al., 2016) and the prototypical pre-Zero-Debris platform. The three differ in payload-to-launcher mass ratio, upper-stage propellant chemistry and disposal scheme, and these differences propagate through the toolchain to regime-specific SSCI signatures.

The reference mission for normalisation is the 500 kg, 700 km sun-synchronous smallsat of the companion engineering paper (Toson, 2026) (Section 3.4).

The functional unit is one mission over its operational life, on a cradle-to-orbital-end-of-life boundary extending the PEFCR4Space definition (European Commission and European Space Agency, 2025) to the orbital operational phase; this per-mission basis compares total mission footprints, not a per-service unit; a delivered-function normalisation (per operational-year or payload capability) is left to future work. The lifecycle inventory of each case study is constructed from public data sources for the terrestrial domain, from launch vehicle propellant chemistry and re-entry oxidation models for the atmospheric domain, and from the Celestrak catalogue snapshot for the orbital domain.

For the terrestrial domain the inventory is characterised through the EF 3.1 method (Fazio et al., 2018) on the SusLifePath database: each product system is built programmatically from its material inventory, with a documented assembly-integration-test electricity overhead, and the EF 3.1 weighted single score is computed through the openLCA bridge of Section 3.5. A data-quality caveat applies. SusLifePath is a purpose-built inventory, not a validated background database such as ecoinvent;

its flows were screened and the dominant ones anchored to published material-level values. In particular the blue-water consumption of the metal and electronics processes was re-grounded to literature intensities, an early version having carried unphysical water inputs that, weighted through the AWARE scarcity model, masked all other categories. After re-grounding, the terrestrial single score is dominated by climate change (~41–44%), with resource use, particulate matter and water use as secondary contributors, the profile expected of electronics-intensive space hardware. The terrestrial scores should be read as order-of-magnitude estimates pending characterisation against a validated background database (future work); the cross-domain conclusions do not depend on terrestrial precision, the terrestrial domain being the smallest of the three for nearly every mission. The launch propellant is characterised only in the atmospheric domain; its cradle-to-gate manufacturing burden is a known omission of the terrestrial inventory (Section 3.1). For the atmospheric domain, the propellant share attributable to each payload follows from the launcher propellant mass and the payload-to-LEO mass ratio: RedPill contributes a negligible fraction of the Falcon 9 load, Sentinel-6 7.5% and ENVISAT 45.6% of the Ariane 5 G+ propellant mass. The characterisation factors are those of Section 3.3, from Ross and Toohey (2019), Maloney et al. (2022), Ferreira et al. (2024) and the WMO 2022 ozone assessment (World Meteorological Organization, 2022). For the orbital domain the three categories are computed against the same Celestrak snapshot used by the companion paper (Toson, 2026).

4.2. SSCI scores and orbital decomposition

Table 3 reports the normalised impact scores per domain and the composite SSCI under the two aggregation strategies and the three weighting scenarios of Section 3.4. All four product systems (three missions and the reference) are computed with the same EF 3.1 pipeline.

Table 3: Domain-specific normalised impact scores and composite SSCI for the three case studies. All scores are dimensionless multiples of the reference mission. The expert-weights row uses the illustrative governance-maturity scenario of Section 3.4.

Quantity	RedPill 2P	Sentinel-6	ENVISAT
$\tilde{I}_T$ (terrestrial)	1.1×10^{-3}	3.84	20.4
$\tilde{I}_A$ (atmospheric)	9.4×10^{-4}	2.43	**380**
$\tilde{I}_O$ (orbital)	**3.18**	1.55	**308**
$SSCI^{lin}$ (equal weights)	1.06	2.60	**236**
$SSCI^{lin}$ (expert weights)	1.11	2.47	265
$SSCI^{geo}$ (geometric mean)	0.015	2.43	134

The scores in Table 3 do not line up monotonically with mass: the PocketCube, four orders of magnitude lighter than the heritage platform, still reaches a composite of order unity, comparable to the reference, because its 550 km shell is among the most congested in LEO and congestion does not scale with mass (Section 3.2). The composite is therefore not a proxy of size.

The relative contributions of the three domains vary in a regime-specific way that the per-sub-category normalisation of Equation (4) makes legible: for the PocketCube the

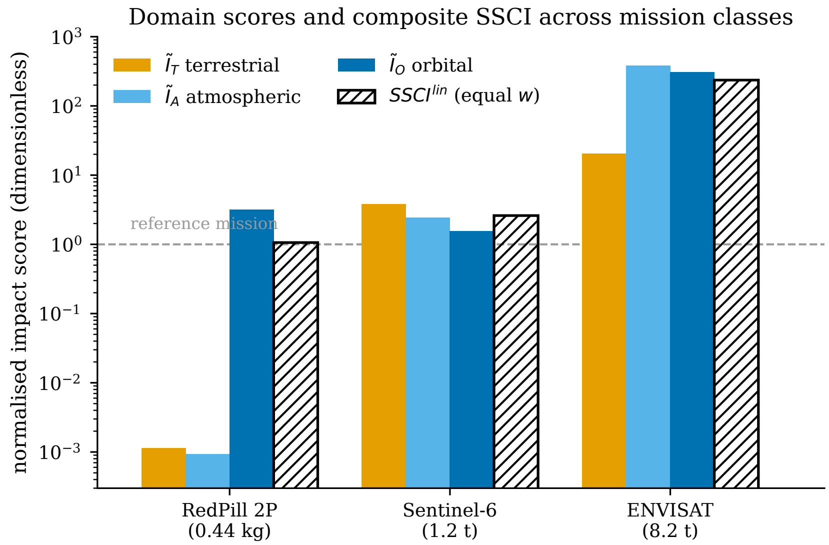

Figure 3: Domain-specific normalised scores and composite SSCI for the three case studies (log scale; the dashed line marks the reference mission). The dominant domain shifts with mission class: orbital congestion for the PocketCube, atmospheric and orbital debris jointly for the heritage platform.

orbital domain dominates and is almost entirely congestion; for the medium smallsat the three domains are within a factor of two and the orbital part is material-criticality-driven (1336 km being too sparse for congestion and too well-disposed for debris); for the heritage platform the atmospheric and orbital domains are comparable (380 and 308 times the reference) and jointly dwarf the terrestrial. The pattern persists across all weighting scenarios.

The ratio $SSCI^{lin}/SSCI^{geo}$ diagnoses how unevenly the impact is distributed across domains: close to unity for the balanced medium smallsat (1.07), 1.8 for the heritage platform, and largest by far for the PocketCube (≈ 70), whose impact is concentrated almost entirely in orbital congestion, so its geometric mean collapses to 1.5×10^{-2} against a linear 1.06. A large ratio flags a mission whose footprint is concentrated in a single domain, itself policy-relevant.

The decomposition of the orbital domain into its three components (*DGP*, *CC*, *MC*; Section 3.2) reveals patterns that are distinctly regime-specific and that single-indicator approaches cannot expose.

Table 4: Decomposition of the orbital impact score $\tilde{I}_O$ into its three normalised components (Equation 4). Each mission class is dominated by a different sub-category.

Component	RedPill 2P	Sentinel-6	ENVISAT
DGP (debris generation)	0%	1%	**98%**
CC (congestion contribution)	**100%**	0%	0%
MC (material criticality)	0%	**99%**	2%

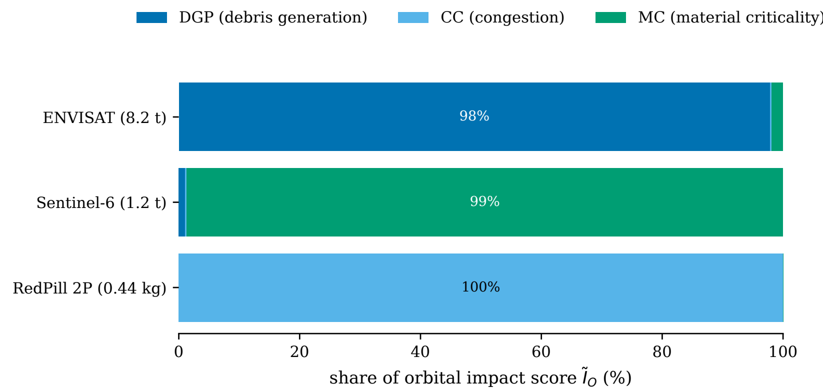

Figure 4: Decomposition of the orbital impact score into its three components (debris generation, congestion contribution, in-orbit material criticality), normalised to 100% per mission. The three mission classes display distinct regime signatures.

Each mission class is dominated by a different orbital sub-category (Table 4), the three together covering the distinct ways a mission burdens the orbital environment. The PocketCube is almost entirely congestion (100%): its 550 km shell coincides with the densely populated Starlink band and, CC being size-independent, a sub-kilogram platform there imposes many times the reference congestion. The medium smallsat at 1336 km sits in a sparse band where debris and congestion are negligible, so its orbital score is material-criticality-driven (99%), amplified by the higher-altitude orbit-loss multiplier. The heritage platform is dominated by debris-generation potential (98%), the defunct eight-tonne case for which ECOB was originally calibrated.

The atmospheric domain decomposes differently. For the heritage platform 99% of the atmospheric score originates from the launch: the two ammonium-perchlorate solid boosters of the Ariane 5 G+ release alumina nanoparticles and chlorine compounds, of which the share attributed to the ENVISAT payload (46% of the booster propellant load, by payload-to-LEO mass ratio) is of order 65 t of alumina and 45 t of chlorine. A terrestrial-only LCA misses this entirely: ENVISAT's normalised atmospheric score is 380 times the reference, against a terrestrial 20.

For the PocketCube and Sentinel-6, both on the Falcon 9 kerosene-LOX architecture, the launch share of the atmospheric impact is 76–81%, the remainder from re-entry demise, proportionally larger for the PocketCube given its assumed full demise.

4.3. Sensitivity and uncertainty

The sensitivity to the weighting is characterised by Dirichlet sampling over the weight simplex with 10^4 samples (random seed fixed in the released code for reproducibility). The fifth-to-ninety-fifth percentile range of $SSCI^{lin}$ is $[0.08, 2.5]$ for the PocketCube, $[1.9, 3.4]$ for Sentinel-6 and $[93, 344]$ for ENVISAT (Figure 5): the heritage platform is separated from the other two by more than an order of magnitude under every admissible weighting. The PocketCube has by far the widest relative spread, a direct expression of its concentrated profile, so the small/medium ordering can invert under extreme weightings while the heritage platform's dominance is weighting-invariant.

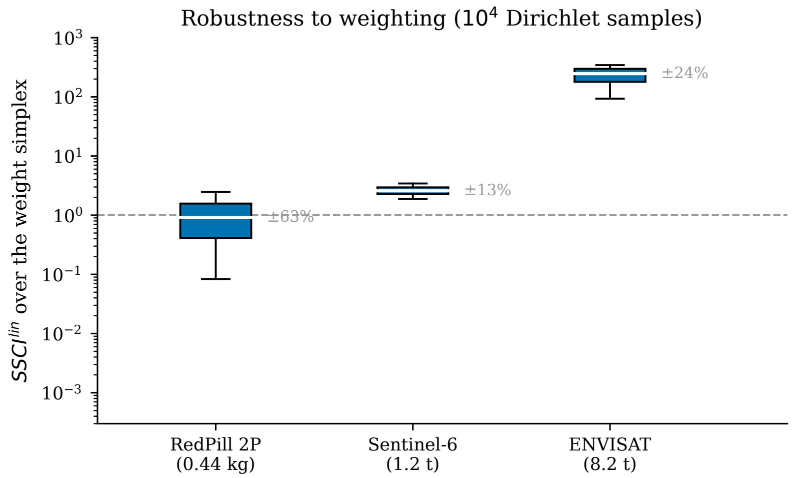

Figure 5: Robustness of the composite to the weighting choice, from 10^4 Dirichlet samples over the weight simplex (log scale; box = interquartile range, whiskers = 5th–95th percentile, dashed line = reference mission). The heritage platform is separated from the other two by over an order of magnitude under every weighting; the PocketCube's wide spread reflects its single-domain (congestion) profile.

A second exercise targets the atmospheric factor the WMO 2022 assessment singles out as unresolved, the black-carbon effectiveness coefficient κ_{BC}, which the assessment reports may be comparable per unit propellant to rocket chlorine (World Meteorological Organization, 2022, Sec. 7.2.8.1), a reading that would raise κ_{BC} by more than an order of magnitude. A one-at-a-time sweep from the baseline ($\kappa\tau_{BC} = 1.8$ yr) to per-propellant-mass parity with chlorine ($\kappa\tau_{BC} \approx 40$ yr, a factor of 22) leaves the mission ranking unchanged. The two kerosene-launched missions are almost invariant, because black carbon scales identically in their numerator and in the kerosene reference denominator. ENVISAT is informative for the opposite reason: black carbon is only ~0.01% of its solid-rocket-motor atmospheric inventory, so its *raw* atmospheric score barely moves; its normalised score falls from 380 to 121 times the reference only because the kerosene reference's own score rises under a larger κ_{BC}. The κ_{BC} sweep therefore probes the reference, not ENVISAT. The factors that actually govern ENVISAT are the residence times of its alumina and chlorine, which carry 65% and 29% of its atmospheric score; a complementary sweep of those two over the literature ranges (alumina $\kappa\tau$ 2.5–7.5 yr, chlorine 2.4–4.8 yr) gives an ENVISAT band of 181 to 380 times the reference, with the ranking preserved in every cell. The central finding is therefore robust to the stratospheric characterisation: the launch-species residence times set the width of the band but not its qualitative conclusion.

Two further checks confirm the robustness of the orbital domain. Varying the orbit-loss multiplier η^{orbit} over a floor of 0.1–0.5 and a lower anchor of 500–700 km with an upper anchor of 1500–2500 km leaves the three case studies' dominant sub-category and the SSCI ranking unchanged; and the dependence of the orbital decomposition on the normalisation reference is quantified in Section 4.5 (Table 6).

4.4. Comparison with single-indicator approaches

The added value of the composite quantification is quantified by comparing the SSCI to the single-indicator alternatives currently in use: the PEFCR4Space terrestrial single score, and the ECOB-derived debris environment rating.

Table 5: Single-indicator scores and the gap of a terrestrial-only (PEFCR-style) indicator relative to $SSCI^{lin}$ (equal weights). A negative gap is an underestimate.

Case study	PEFCR-style	SSCI orbital	$SSCI^{lin}$	Gap
RedPill 2P	1.1×10^{-3}	3.18	1.06	**−100**%
Sentinel-6	3.84	1.55	2.60	+47%
ENVISAT	20.4	308	**236**	**−91**%

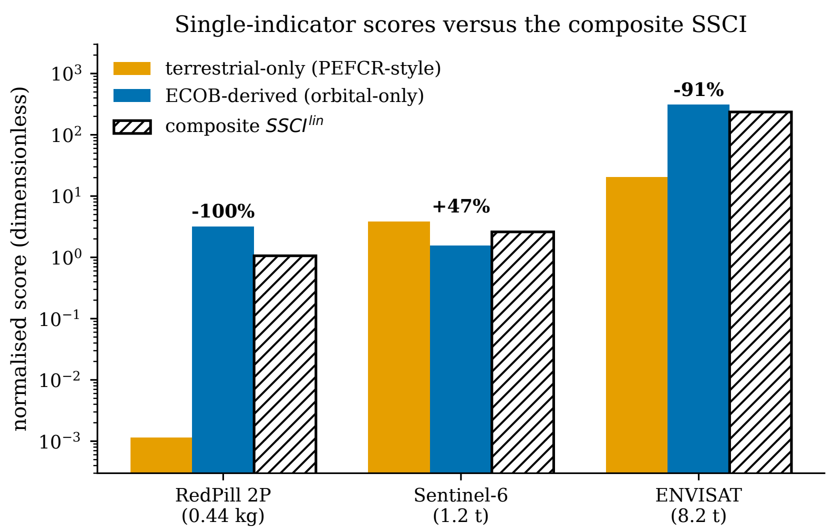

Figure 6: Single-indicator scores (terrestrial-only PEFCR-style; the SSCI orbital sub-score) against the composite SSCI, with the terrestrial-only gap annotated (log scale). A terrestrial-only indicator underestimates the PocketCube and the heritage platform by about 100% and 91% and overestimates the medium smallsat by 47%.

The terrestrial-only score, what a PEFCR4Space-style assessment would report, mis-estimates the composite in both directions and by large factors. It overestimates the medium smallsat by 47% (the materially intensive bus exceeds its modest atmospheric and orbital scores), underestimates the heritage platform by 91% (PEFCR4Space characterises neither the solid-rocket-motor emissions nor the debris-generation potential), and underestimates the PocketCube by essentially 100%, its footprint being almost entirely orbital congestion, invisible to any terrestrial indicator (Figure 6). The second column reports the SSCI orbital sub-score, which captures the orbital impact but, used alone, would miss the terrestrial and atmospheric phases.

Only the SSCI spans all domains; it captures trade-offs no single-indicator approach can express. Section 4.5 tests whether this holds across the operational population.

4.5. Catalogue-scale application

Three case studies establish the regimes but not their prevalence. To test whether the pilot's conclusions hold across the population, the framework is applied to 300 missions whose altitudes are drawn from the real active-LEO distribution of the May 2026 Celestrak catalogue (14 727 objects, median 482 km), with mass, material inventory and launch vehicle assigned by mission-class archetype, using an assumed illustrative class mix (50% CubeSat, 35% smallsat, 12% medium, 3% large) to stand in for the mass the public catalogue does not carry. The population facts reported below depend only on the measured altitude distribution, not on this mix, which enters only the mass-coupled terms. The debris component is computed through a single batch of the ECOB-proxy module, the congestion contribution from the same density model as the case studies, and the atmospheric and terrestrial domains from the archetype assignment. Mass, materials and launcher remain archetype-assigned, so the exercise is an altitude-distribution sensitivity over four mass classes, not a per-object inventory of the population; its purpose is to test whether the pilot's regime structure persists once the real altitude distribution, not three hand-chosen points, drives the orbital terms.

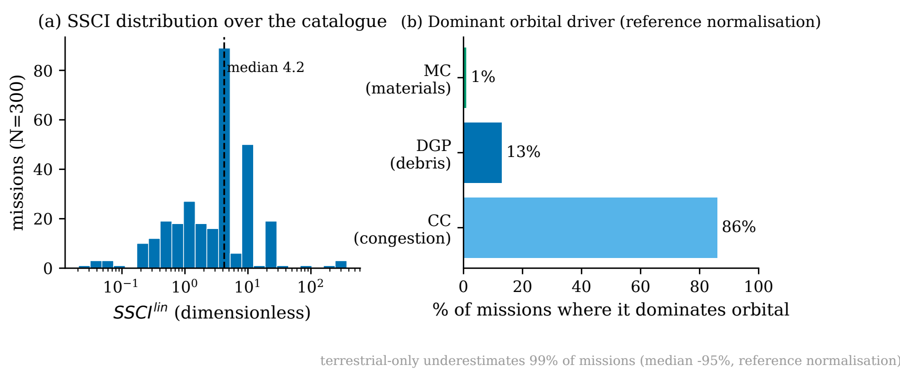

Figure 7: Catalogue-scale application to $N = 300$ missions with altitudes from the real active-LEO distribution. (a) Distribution of the composite SSCI. (b) Share of missions for which each orbital sub-category is the dominant one under the reference-mission normalisation; this share is reference-dependent (Table 6), whereas the concentration of the active population in the dense 400–600 km shells is not.

Two findings follow (Figure 7). The first needs no normalisation and no sampling: across the full catalogue of 14 727 active objects, 86% operate below 600 km and 97% occupy shells denser than the 700 km reference orbit, so the typical mission's absolute congestion exposure (satellite-years within protected conjunction volumes) far exceeds the reference's, a burden terrestrial Life Cycle Assessment does not register. The active LEO population is concentrated where operational congestion is highest.

That concentration is a fact about altitudes. Turning it into a single dominant orbital *sub-category*, however, requires a choice of normalisation, and there the answer is less robust. Under the reference-mission normalisation of Equation (4) congestion dominates for 86% of the sample, but this share falls to 19–46% under population-based normalisations (Table 6), because the sparse 700 km reference carries a near-zero congestion value that inflates every other mission's normalised congestion. Invariant across all schemes are the per-mission rankings of the three case studies (congestion for the PocketCube, material criticality for the medium smallsat, debris generation for the heritage platform) and the non-redundancy of the three sub-categories. The population share attributed to congestion therefore depends on the choice of reference.

Table 6: Robustness of the catalogue-scale orbital decomposition to the normalisation: share of the $N = 300$ sample for which each orbital sub-category is dominant, under four schemes. The reference-mission scheme used elsewhere is the most congestion-weighted; population-based schemes are not, yet the three case studies keep their dominant sub-category under every scheme.

Normalisation	DGP	CC	MC
Reference mission (baseline)	13%	**86%**	1%
Catalogue median	43%	28%	29%
Catalogue geometric mean	39%	46%	15%
Congested-shell reference	41%	19%	40%

A terrestrial-only PEFCR4Space-style indicator underestimates the composite for the large majority of the sample, omitting the orbital and stratospheric burdens entirely: under the reference normalisation 99% of missions are underestimated, by a median of 95%. The negative *sign* follows structurally from the terrestrial domain being the smallest of the three for nearly every mission, so the catalogue result quantifies how large this built-in direction is; that magnitude inherits the reference-dependence of the orbital normalisation. The composite spans more than four orders of magnitude across the sample, but its interquartile range is under one decade (median 4.2, IQR $[1.2, 8.6]$), so the dynamic range is carried by the heavy, defunct tail, not the operational bulk.

5. Discussion

5.1. Methodological contributions and limitations

The framework carries three main limitations. The orbital categories rest on debris-evolution models whose predictive accuracy beyond one or two centuries is uncertain, and the characterisation factors will require periodic recalibration as models and observations improve. In the pilot the debris component is supplied by the snapshot proxy of Section 3.2 rather than a full ECOB/MASTER-class run, so the orbital scores are proxy-based: the validation indicates the mission *ranking* is preserved, though absolute levels would shift under a full debris-evolution computation. The atmospheric modelling does not yet capture all chemical pathways relevant to ozone depletion and aerosol loading. And the weighting strategies, though diverse, do not exhaust the legitimate aggregation philosophies; the choice between linear and geometric-mean aggregation reflects a value judgement on which the framework offers two formulations rather than a single answer.

5.2. Regulatory interfaces

The SSCI interfaces, as a technical reference rather than a legal instrument, with the methodology now consolidating European space-environmental governance. The most concrete interface is the PEFCR4Space draft (European Commission and European Space Agency, 2025), which will become the reference methodology for environmental certification of space products in the European Union but whose current treatment of the orbital and atmospheric phases does not reflect the maturity of the corresponding literatures; because the SSCI is compatible with the EF 3.1 method, its congestion, material-criticality and stratospheric terms are candidates for consideration, in a subsequent revision, as supplementary indicators for missions whose footprint is dominated by those phases, without restructuring the framework. The *DGP* and *CC* categories likewise offer quantitative metrics for the compliance objectives of debris-mitigation guidelines (Inter-Agency Space Debris Coordination Committee, 2021) and the ESA Zero Debris Charter (European Space Agency, 2023). Whether and how these soft-law instruments, the EU Ecodesign for Sustainable Products Regulation (European Parliament and Council, 2024), or the forthcoming EU Space Law would adopt such an indicator is a governance question beyond the scope of this work.

5.3. Practical implications for eco-design and mission planning

Beyond regulation, the SSCI supports concrete design decisions: decomposing the composite into domain-specific contributions lets designers identify, early in architecture definition, which domain drives the footprint and direct effort accordingly. The dominant domain depends on mission class (Section 4); for the PocketCube it is orbital congestion, where the eco-design signal points toward orbit selection over mass reduction; it is invisible to single-indicator approaches.

The most consequential lever is the chemistry of the launch vehicle. The atmospheric domain is dominated, in absolute terms, by the presence or absence of solid rocket motors: in the pilot the per-tonne-of-payload stratospheric impact of the APCP-boosted Ariane 5 G+ exceeds that of the kerosene-LOX Falcon 9 by a factor of about 28, because ammonium-perchlorate propellant releases alumina and chlorine species with stratospheric residence times of years to decades (Ferreira et al., 2024) whereas kerosene-LOX releases mainly black carbon and water vapour with lower ozone-depletion potential. For environmentally sensitive payloads the choice between an SRM and a non-SRM launcher is thus a first-order driver of the footprint, one that current debris-mitigation guidelines and the PEFCR4Space draft do not differentiate. The SSCI makes it quantifiable at the early-design stage, and the κ_{BC} sensitivity of Section 4.3 confirms the ranking by launcher chemistry is stable.

6. Conclusions

The expansion of space activities generates environmental pressures across the terrestrial, atmospheric and orbital domains, each currently addressed by a separate methodological tradition. This paper has presented the Space Sustainability Composite Indicator (SSCI), a composite sustainability indicator that integrates the three within a single quantification: it couples terrestrial Life Cycle Assessment with an ECOB-aligned debris component and two complementary categories (operational congestion and in-orbit material criticality), compares two aggregation rules, and is realised as an open toolchain releasing a 54-process inventory library, one of the few openly released inventory libraries for the Life Cycle Assessment of space systems.

Applied to the three case studies and to a 300-mission catalogue-scale sample, the framework shows four properties that distinguish it from single-indicator approaches. The composite is not a proxy of size: the sub-kilogram PocketCube reaches a composite of order unity because congestion does not scale with mass and its 550 km shell is among the most contested in LEO. The three orbital sub-categories are regime-specific, each governing a distinct mission class (Table 4). Scaling to the real active-LEO population shows that most missions occupy shells far denser than a representative reference orbit, so the absolute congestion they impose is a first-order orbital burden invisible to terrestrial assessment; a terrestrial-only PEFCR4Space-style indicator consequently underestimates the multi-domain footprint for the large majority of missions, although the population share attributed to congestion is reference-dependent (Section 4.5). And the ratio between the two aggregations diagnoses how unevenly a mission's impact is distributed. Together these support the central message that the footprint of a space mission is governed by where and how it operates as much as by what it is built from, and that congestion of contested shells, a function of presence rather than mass, is invisible to terrestrial-only assessment.

Several directions of future research follow. The extension to constellation- and fleet-level analysis, where the interaction between individual missions and the collective orbital environment becomes a first-order phenomenon, is a natural next step. The structured calibration of the weights through stakeholder elicitation would establish a reference weighting reflecting documented societal preferences. And the integration of the SSCI with prospective LCA would support anticipatory assessment of emerging launch and propulsion technologies. Each is enabled by the open implementation and the SusLifePath inventory library.

The open toolchain and the SusLifePath inventory library are released so that other groups can apply the SSCI to their own missions and build on it.

CRediT authorship contribution statement

Federico Toson: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The SSCI toolchain (Python orchestrator, domain modules, MATLAB bridge scripts and mission descriptors) and the SusLifePath OpenLCA process library are released under MIT licence and deposited on Zenodo for citable reference at submission. Static result artefacts (per-mission `results.json`, the EF 3.1 weighted terrestrial breakdown and the sensitivity outputs) are included so that the orbital, atmospheric and aggregation results reproduce without an OpenLCA installation; the terrestrial single score itself requires the OpenLCA database (the per-category breakdown cached in `results.json` is characterised, not weighted; the weighted profile is in `terrestrial_weighted.csv`). In the released code the geometric-mean aggregation carries the key `SSCI_risk` (= $SSCI^{geo}$).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted tools to support language refinement and code testing. After using these tools, the author reviewed and edited the content and takes full responsibility for the content of the publication.

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